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```
import nltk
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer, WordNetLemmatizer
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')
nltk.download('punkt_tab')
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data] Package wordnet is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Package punkt_tab is already up-to-date!
True
```

1.Tokenization = Word tokenization is the most commonly used method where text is divided into individual words

```
text = " It is obvious that people want to get the best services for the best price. You should also keep in mind that bigger price is better."
print(f"ORIGINAL TEXT:\n{text}\n")
tokens = word_tokenize(text)
print("1. TOKENIZATION (Splitting words):")
print(tokens, "\n")
```

```
['It', 'is', 'obvious', 'that', 'people', 'want', 'to', 'get', 'the', 'best', 'services', 'for', 'the', 'best', 'price', 'also', 'keep', 'in', 'mind', 'that', 'bigger', 'price', 'is', 'better']
```

2.Removing The Stopwords

```
stop_words = set(stopwords.words('english'))
filtered_tokens = [word for word in tokens if word.lower() not in stop_words and word.isalpha()]
print("2. STOPWORDS REMOVAL (Filtering words):")
print(filtered_tokens, "\n")
```

```
['obvious', 'people', 'want', 'get', 'best', 'services', 'best', 'price', 'also', 'keep', 'mind', 'bigger', 'price', 'is', 'better']
```

3. Stemming = Root Word

```
stemmer = PorterStemmer()
stemmed_words = [stemmer.stem(word) for word in filtered_tokens]
print("3. STEMMING :")
print(stemmed_words, "\n")
```

```
3. STEMMING :
['obviou', 'peopl', 'want', 'get', 'best', 'servic', 'best', 'price', 'also', 'keep', 'mind',
```

4.Lemmatization = Convert Into Dict Words

```
lemmatizer = WordNetLemmatizer()
lemmatized_words = [lemmatizer.lemmatize(word) for word in filtered_tokens]
print("4. LEMMATIZATION (Converting to real dictionary base words):")
print(lemmatized_words, "\n")
```

```
4. LEMMATIZATION (Converting to real dictionary base words):
['obvious', 'people', 'want', 'get', 'best', 'service', 'best', 'price', 'also', 'keep', 'min
```

```
translator = str.maketrans('', '', string.punctuation)
text_without_punctuation_string = ' '.join(tokens).translate(translator)
punctuation_removal = word_tokenize(text_without_punctuation_string)
print("5. PUNCTUATION REMOVAL:")
print(punctuation_removal, "\n")
```

```
PUNCTUATION REMOVAL:
['is', 'obvious', 'that', 'people', 'want', 'to', 'get', 'the', 'best', 'services', 'for', 'the
```